TokTickIT — Lab 4 Submission

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Verification (on `main`): **server 46/46 tests**, **client 18/18 tests**, `tsc --noEmit` clean, `vite build` OK.

Answer Part 1 — Git Use with Engineering Workflow (10)

Branch flow: each issue on a `feature/4.x-*` branch → PR into `lab4-staging` → `lab4-staging` merged into `main`. Every feature PR was reviewed and approved by the peer (SaintCrois) before merge.

PR	Issue	Branch	Reviewer
#56	4.1 Spec	feature/4.1-sprint4-specs	Approved — SaintCrois
#57	4.2 Actions Taken backend	feature/4.2-actions-taken-foundation	Approved — SaintCrois
#58	4.4 Ticket workflow	feature/4.4-ticket-workflow	Approved — SaintCrois
#59	4.5 Dashboards backend	feature/4.5-role-dashboards	Approved — SaintCrois
#60	4.3 Actions Taken UI	feature/4.3-actions-taken-ui	Approved — SaintCrois
#61	4.6 Responsive + hardening	feature/4.6-hardening-responsive	Approved — SaintCrois
#62/#63/#65	integration / dashboard UI / main	→ lab4-staging → main	Merged

Issues #44–#49 (GitHub Issues) cover specification, Actions Taken, workflow, dashboards, UI, and hardening. Rendered: `docs/lab-04/reviewer.md`. README, `.gitignore`, and the repository directory structure are current on `main`.

Answer Part 2 — Spec DD (5)

Rendered: `docs/lab-04/specification.md`. Contains numbered Functional Requirements (FR-01+), Business Rules (BR-01+ incl. the Actions Taken rules and the full Ticket status-transition matrix BR-09), the IT-Staff resolution gate (BR-08), dashboard calculation rules (BR-10), two justified data-

design decisions, acceptance criteria (AC-01+), and the Product Definition of Done. The spec was committed in PR #56 before the implementation PRs, evidencing spec-before-code.

Answer Part 3 — Test DD and Traceability (10)

Rendered: `docs/lab-04/tests.md` (traceability matrix mapping AC → test file). Passing suites on `main`:

```
server:  Test Files 13 passed (13)   Tests 46 passed (46)
client:  Test Files 10 passed (10)   Tests 18 passed | 2 todo (20)
tsc --noEmit: clean      vite build: OK
```

Test files: `server/tests/lab-04/{actions-taken,ticket-workflow,requester-dashboard,staff-dashboard}.api.test.ts`; `client/tests/lab-04/{ActionsTaken,RequesterDashboard,StaffDashboard,Responsive}.test.tsx`.

Answer Part 4 — AI Use with Reflection (5)

Rendered: `docs/lab-04/ai-use.md` — tools named, 8 key engineering prompts with verifications, and a brief reflection.

Answer Part 5 — Working IT Staff Dashboard UI (5)

Backend: `GET /api/dashboard/staff` (unassigned, owned-by-me, by-status, by-priority, recently-updated — all computed server-side, verified in `staff-dashboard.api.test.ts`). Frontend: `client/src/Dashboard.tsx`, reachable from the **Dashboard** nav item; metric cards drill through to the Staff Queue. [Insert screenshot: Staff Dashboard — desktop]

Answer Part 6 — Working Actions Taken UI (10)

Backend: `GET/POST/PATCH /api/tickets/:id/actions` with role enforcement (staff/admin write, requester read-own) and follow-up-note validation. Frontend: `client/src/ActionsTaken.tsx` on the Ticket Detail screen — list (oldest-first), create and edit forms, follow-up note required when flagged, duplicate-submit guard, requester read-only. [Insert screenshot: Ticket Detail with multiple Actions Taken + create form]

Answer Part 7 — Working Ticket Workflow (5)

Full 8-status transition matrix (New, Open, In Progress, Waiting for Requester, Resolved, Closed, Reopened, Cancelled) enforced on the backend; illegal transitions → 400; stale updates → 409; a Requester "appears resolved" signal is advisory and never sets Resolved. Append-only, stable ordering. Verified in `ticket-workflow.api.test.ts`. [Insert screenshot: status control showing only permitted transitions]

Answer Part 8 — Requester Dashboard & Final Regression (5)

Requester dashboard (`GET /api/dashboard/requester`) is owner-scoped: total open, waiting-for-me, recently updated, recently resolved — ownership enforced server-side (verified in `requester-dashboard.api.test.ts`). Regression: all Lab 1–3 suites pass on `main` . [Insert screenshots: Requester Dashboard + representative regression screens]

Answer Part 9 — Zen Green UI, Responsive, Accessibility, Final Polish (5)

Rendered: `docs/lab-04/ui-spec.md` . Root-cause fix: the app previously ran a stale compiled `App.js` , so UI edits never shipped; now `main.tsx` and tests load `App.tsx` , `tsconfig` `noEmit:true` prevents re-emitting `.js` . Responsive CSS adds ≤768px / ≤1024px breakpoints, table wrappers, full-width inputs (no iOS zoom), wrapping nav, and `:focus-visible` outlines; no horizontal page scroll. [Insert screenshots: desktop / tablet / mobile for major Lab 4 screens]

The submitted repository and final `main` branch remain the source of truth. Screenshot placeholders marked above should be filled from the running app before final submission.